

Transgenesis: methods and applications

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Transgenesis: definition and function

Transgenic animal : an animal whose genome has been modified

- by the addition of new DNA sequences with random integration (classical transgenesis).
- by the modification of endogenous sequences, thanks to homologous recombination (knock-out, knock-in).

In both cases, these changes are transmissible through the germline, from one generation to the next.

Transgenesis: definition and function

Transgenesis allows to study gene function:

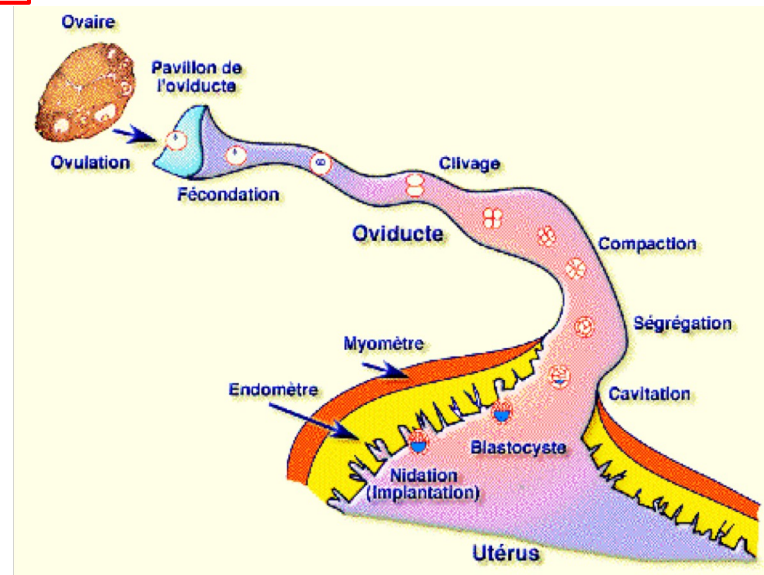
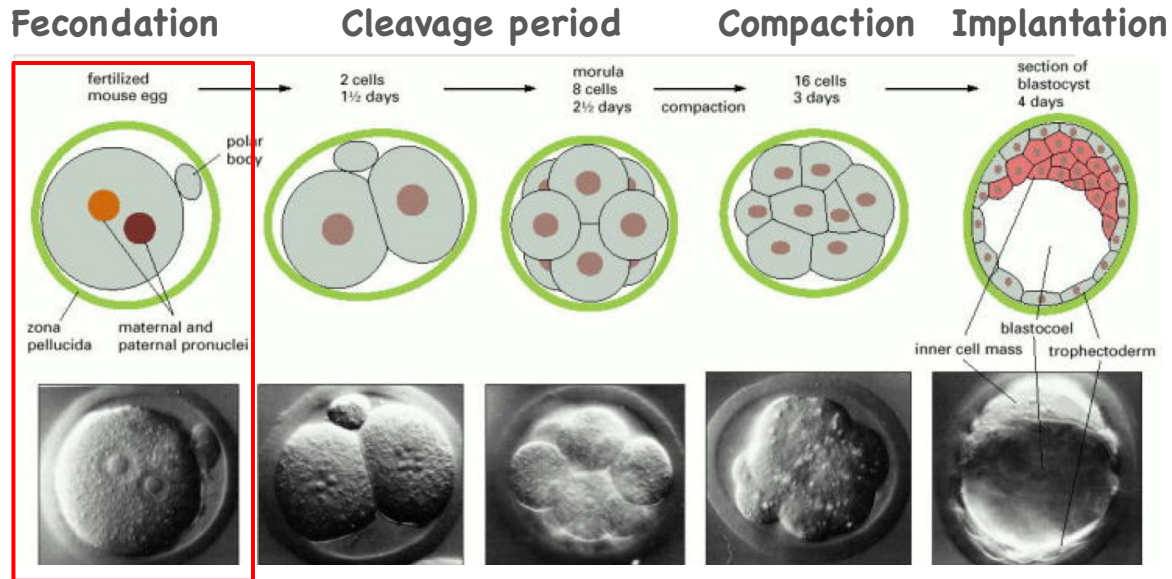
- during embryonic development and organogenesis
- in adults
- in disease models

Classical transgenesis : random integration of the transgene into the host genome

The aim is to introduce one or more exogenous sequences into the genome of the host animal in a stable (transmissible) manner.

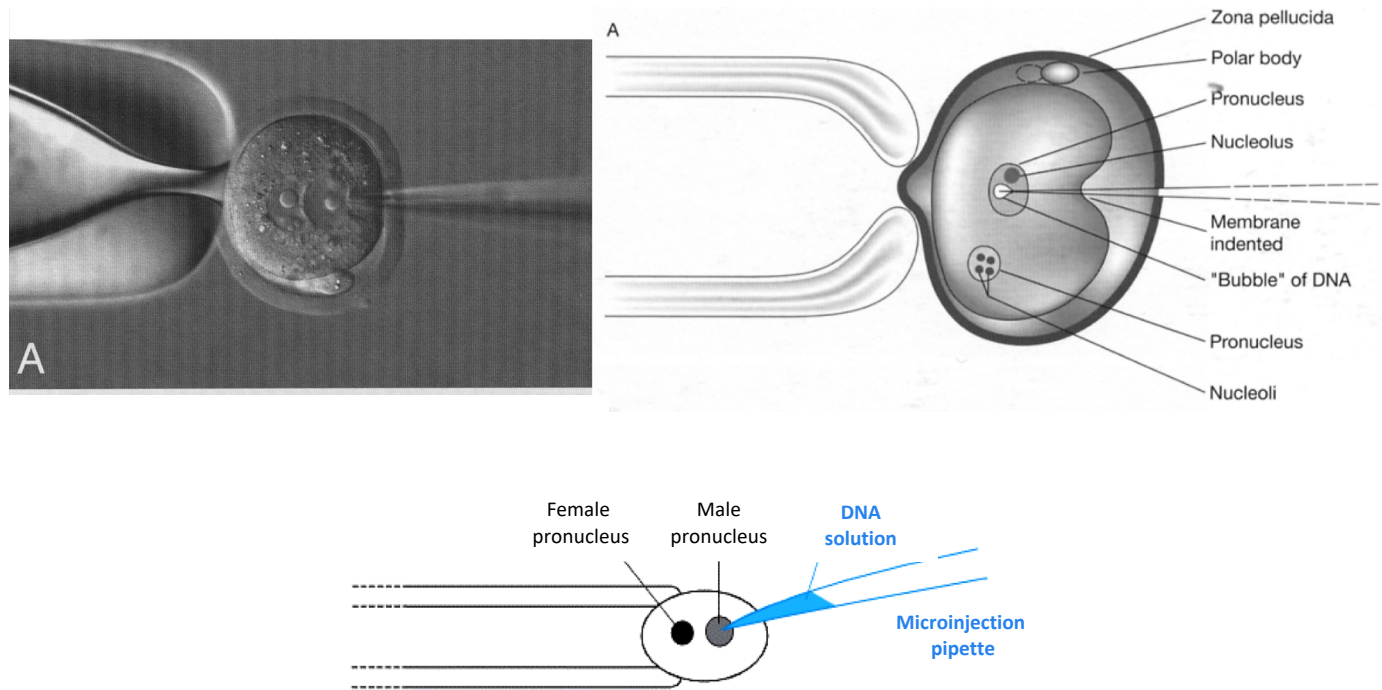
DNA introduction occurs in the embryo at the 1-cell stage, before the fusion of the pronuclei.

Classical transgenesis : random integration of the transgene into the host genome

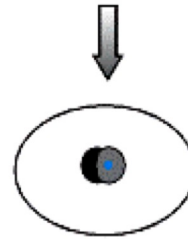
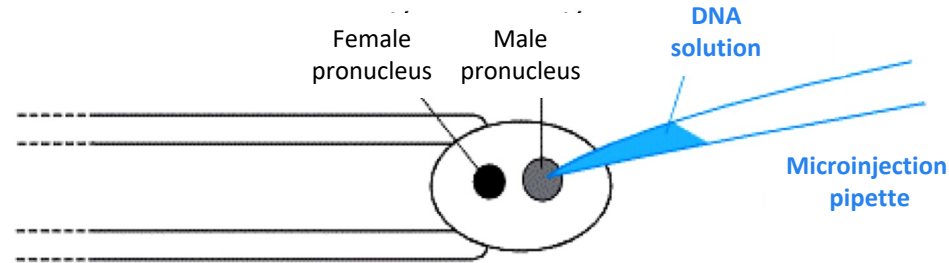


Classical transgenesis : random integration of the transgene into the host genome

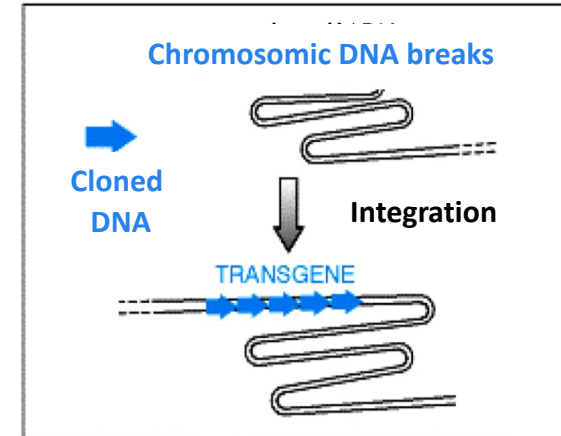
DNA micro-injection into male pronucleus



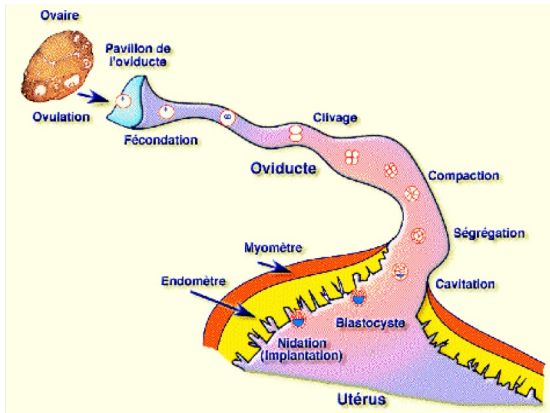
Classical transgenesis : random integration of the transgene into the host genome



Transfert into the oviduct of pseudo-gestant females



Transgenic pup
(the transgene is present in all cells)



Classical transgenesis : random integration of the transgene into the host genome

- Hormonal treatment of female mice to induce a super-ovulation.
- Breeding of treated females.
- The day after: embryos selection at 1-cell stage.
- Micro-injection of linearized DNA into males pronuclei (n=300)
- Immediate transfert of micro-injected eggs into females oviducts of pseudo-gestant female mices.

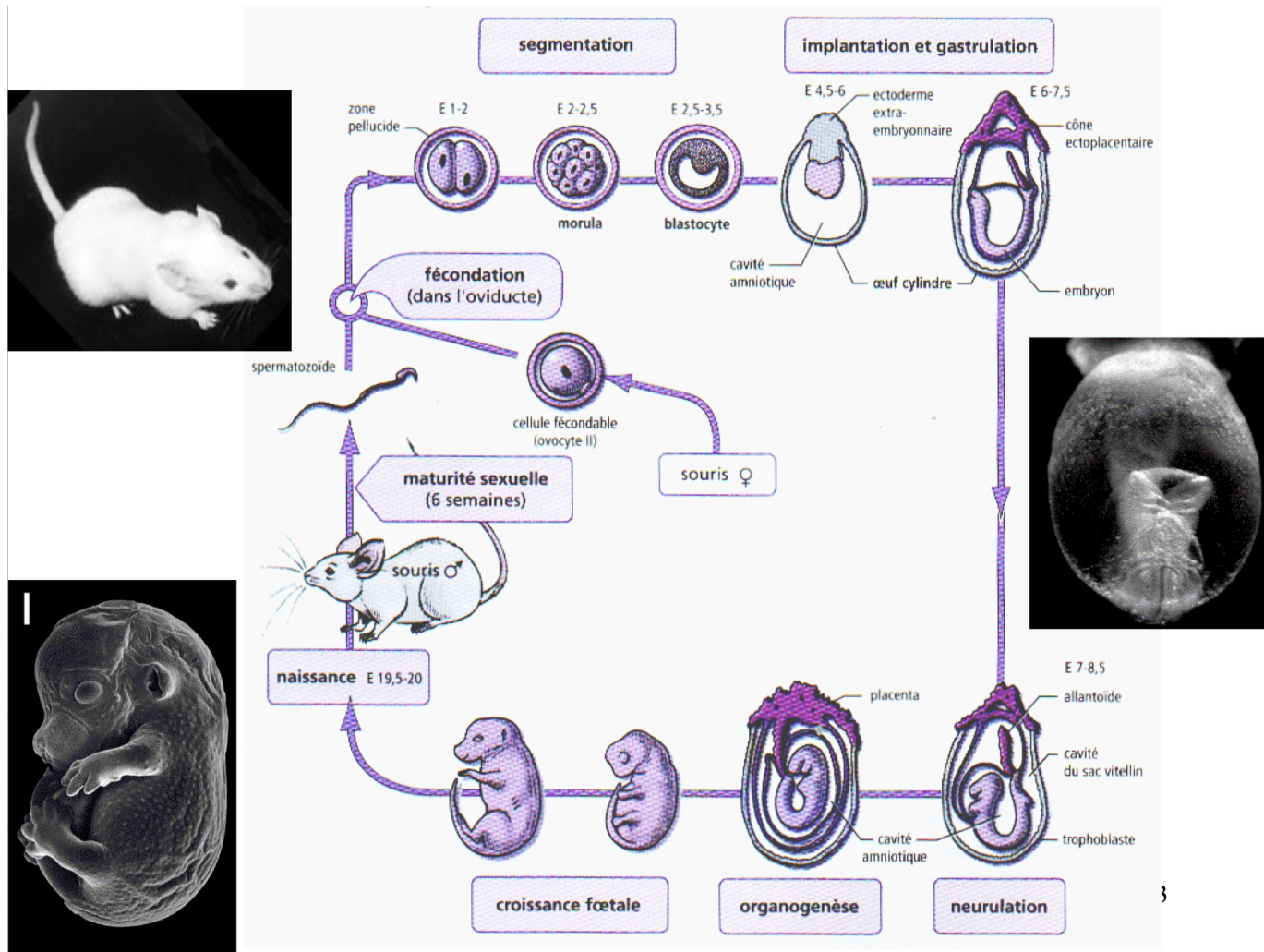
Classical transgenesis

Embryonic analysis during development : **transitory transgenesis**

- (1) Termination of gestation and embryos removal.
- (2) Genotyping of embryos to identify those having integrated the transgene into their genome (30%).
- (3) Phenotypic analysis.

Classical transgenesis

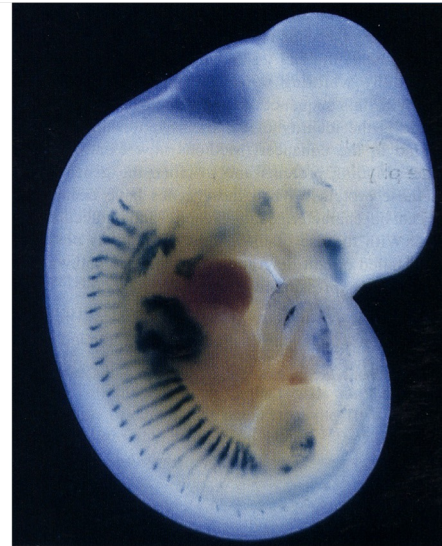
Transient transgenesis : gene regulatory regions analysis during development



Classical transgenesis

Transient transgenesis : gene regulatory regions analysis during development

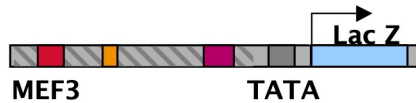
Regulation of *Myogenin* gene



Hybridation in situ
Myogenin E11.5

Classical transgenesis

Transient transgenesis : gene regulatory regions analysis during development



Minimal promoter
Myogenin gene

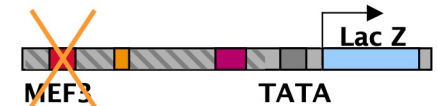


E11.5

in situ hybridization
Myogenin



E11.5



MEF3 binding site mutation

Myog(mut)-LacZ



E11.5

Classical transgenesis

Embryonic analysis during development : **transient transgenesis**

- (1) Termination of gestation and embryos removal.
- (2) Genotyping of embryos to identify those having integrated the transgene into their genome (30%).
- (3) Phenotypic analysis.

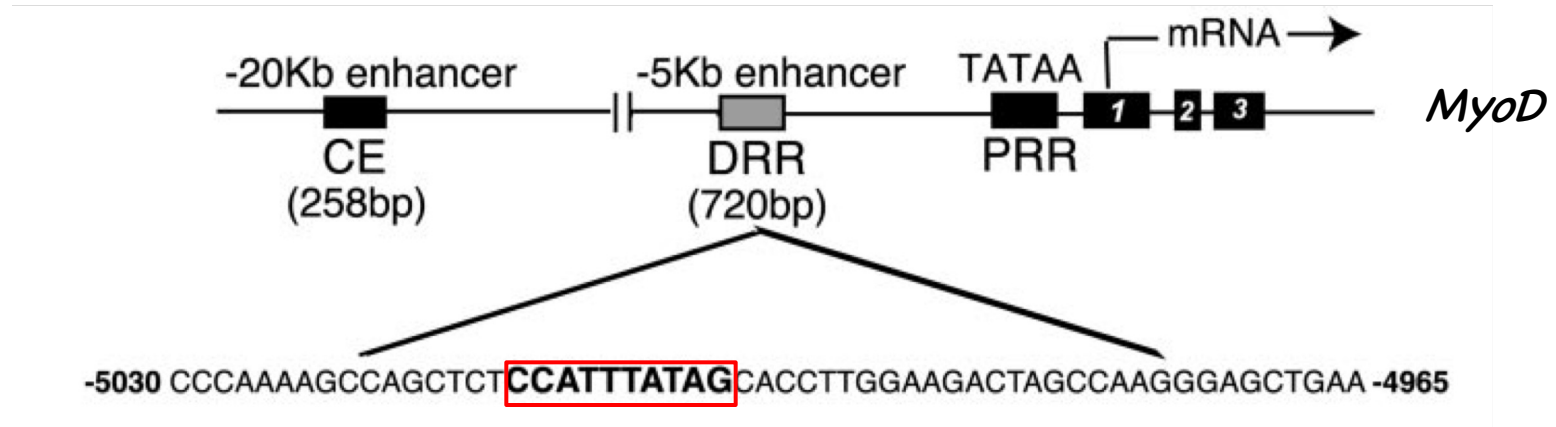
OR

Birth of mice litter (19 to 20 days) : **stable transgenesis**

- (1) Newborn genotyping to identify those having integrated the transgene into their genome (30% of founders)
- (2) Founders (F0) breeding with wild-type animals to establish the mice line.

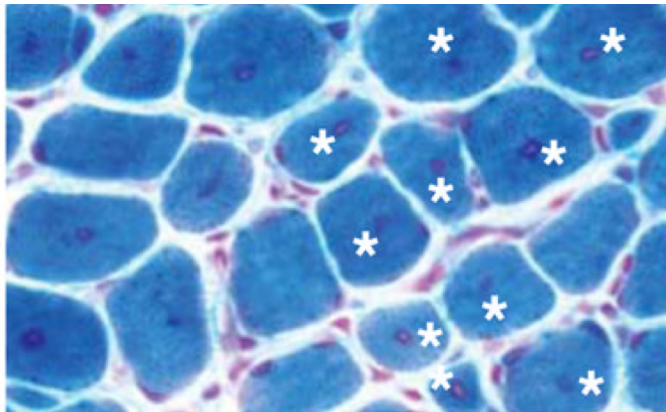
Classical transgenesis

Stable transgenesis: gene regulatory regions analysis in the adult

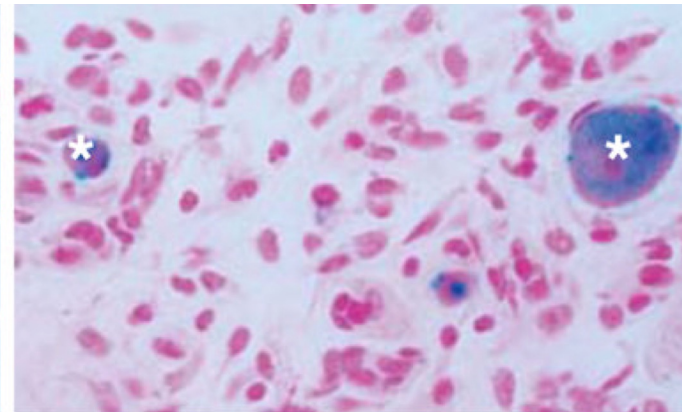


Xgal staining

DRR-PRR-LacZ



DRR(mut)-PRR-LacZ



Regenerating skeletal muscle after injury

Classical transgenesis

Stable transgenesis : consequences of mutated protein surexpression

Pitx2 is expressed in the cornea. The human T68P mutation is associated with Rieger syndrome (cataract formation due to anterior segment dysgenesis).

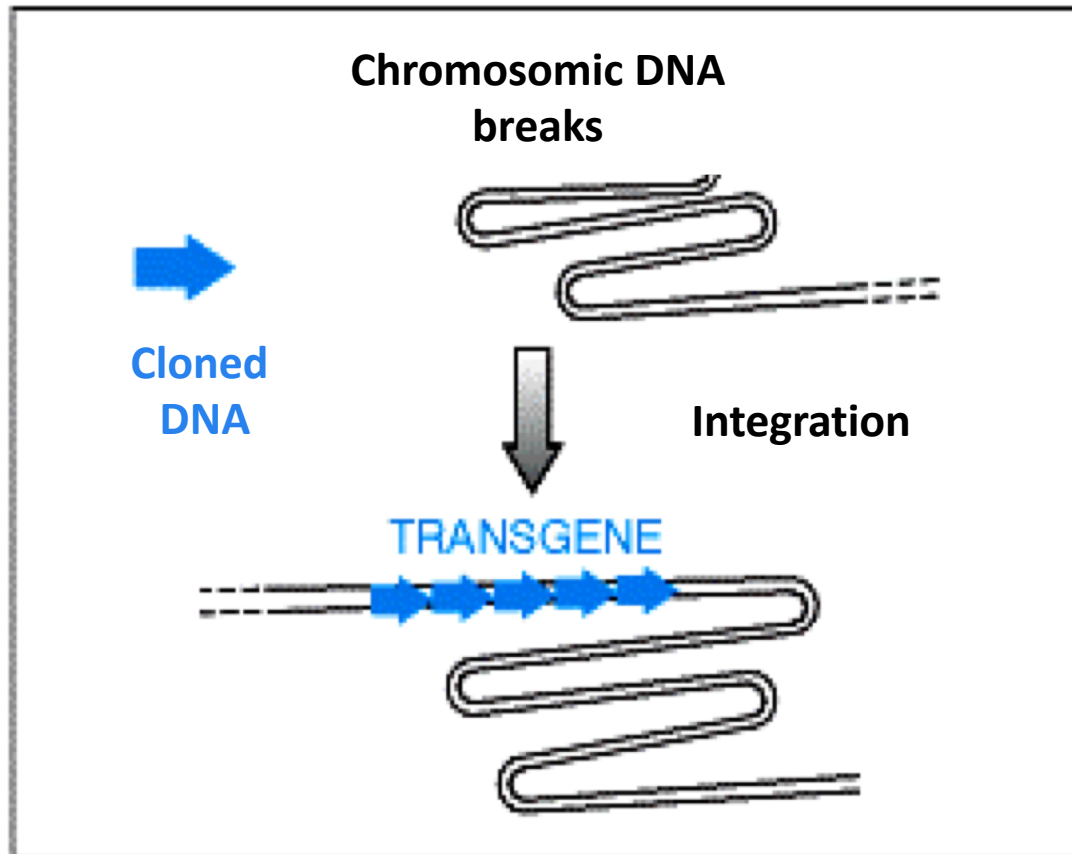
Production of two transgenic lines allowing the expression of Pitx2wt and its mutant Pitx2T68P in the cornea (*Ktcr*: regulatory regions of the *Keratocan* gene which is expressed in the cornea), and analysis of the animal corneas.

Classical transgenesis : random integration of the transgene into the host genome

What kind of transgene ?

- * Plasmid (<20 kb)
- * BAC bacterial artificial chromosome 100–300 kb
- * YAC yeast artificial chromosome 150–2000 kb

Classical transgenesis : random integration of the transgene into the host genome



Classical transgenesis : random integration of the transgene into the host genome

What kind of transgene

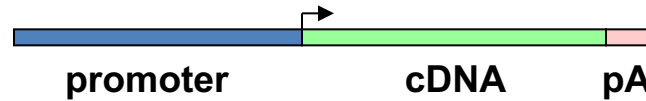
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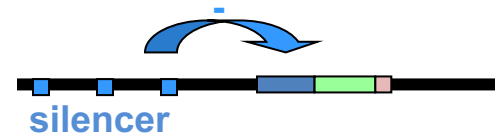
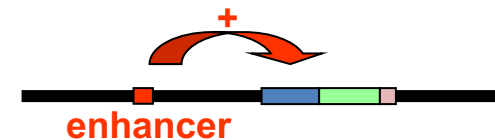
Classical transgenesis : random integration of the transgene into the host genome

* Plasmid (<20 kb)



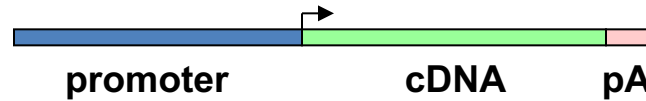
The hazards of integration

- Endogenous gene silencing
- Transgene ectopic activation
- Transgene silencing



Classical transgenesis : random integration of the transgene into the host genome

* Plasmid (<20 kb)



The hazards of integration

- Insertion of multiples copies at a random site in the genome.
- This random insertion can produce different phenotypes from one animal to another.
- It is mandatory to analyze several independant transgenic lines (with n which depends of the studied state) to validate the specificity of the obtained phenotype.

Classical transgenesis : random integration of the transgene into the host genome

* Plasmid (<20 kb)

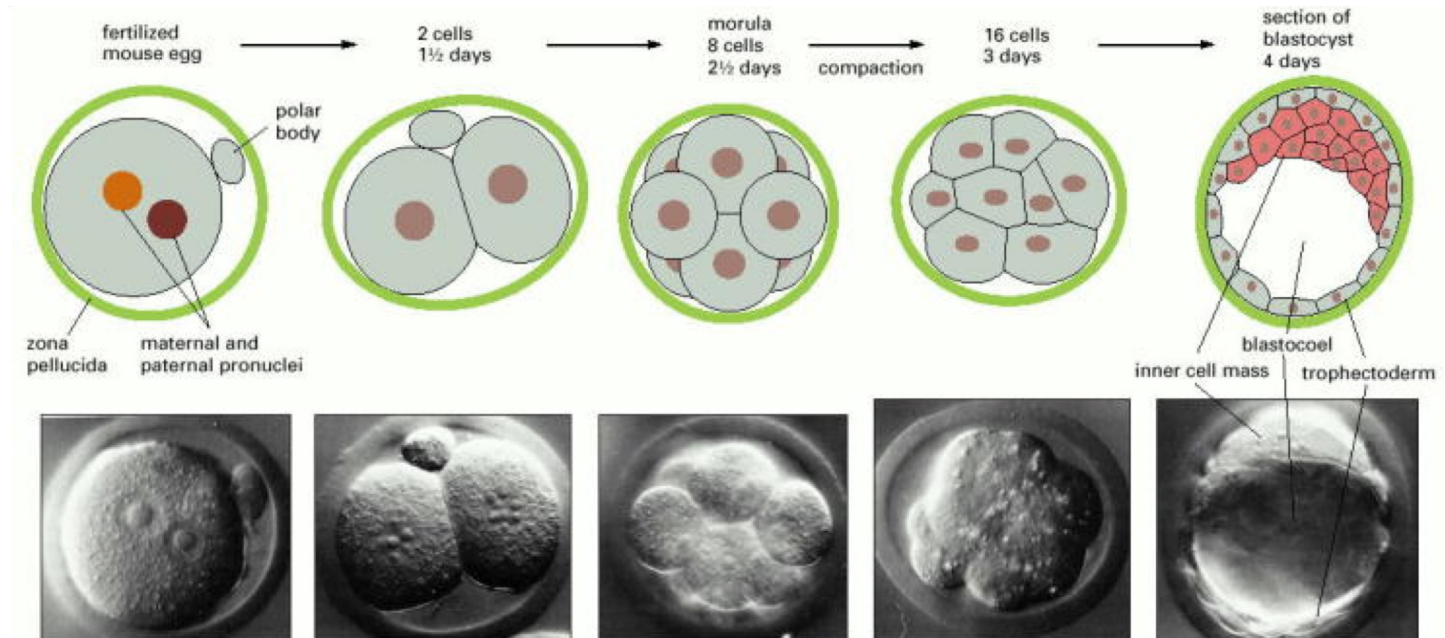
* BAC bacterial artificial chromosome 100–300 kb

* YAC yeast artificial chromosome 150–2000 kb

Allows cloning of the entire locus: introns and regulatory sequences

- Production of all locus products in proportions that reflect the endogenous situation.
- Transgene expression is copy number dependent: analysis of the "gene dosage" effect.
- Transgene expression is independent of the insertion position.

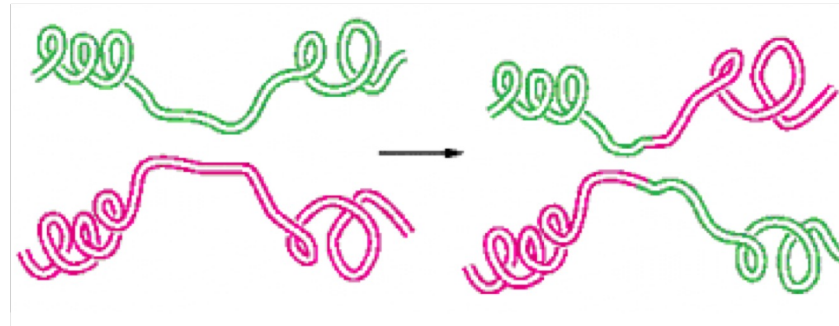
Classical transgenesis : random integration of the transgene into the host genome



DNA Micro-injection
Additional transgenesis
Random insertion
Transitory or stable
transgenic lines
(3 months, 4000 euros)

Homologous recombination : targeted integration of the transgene into the host genome

In presence of homologous DNA sequences, double-strand breaks of the two DNA molecules occur, whose ends re-join with opposite strands



General Recombination:

All homologous sequences can be recombined

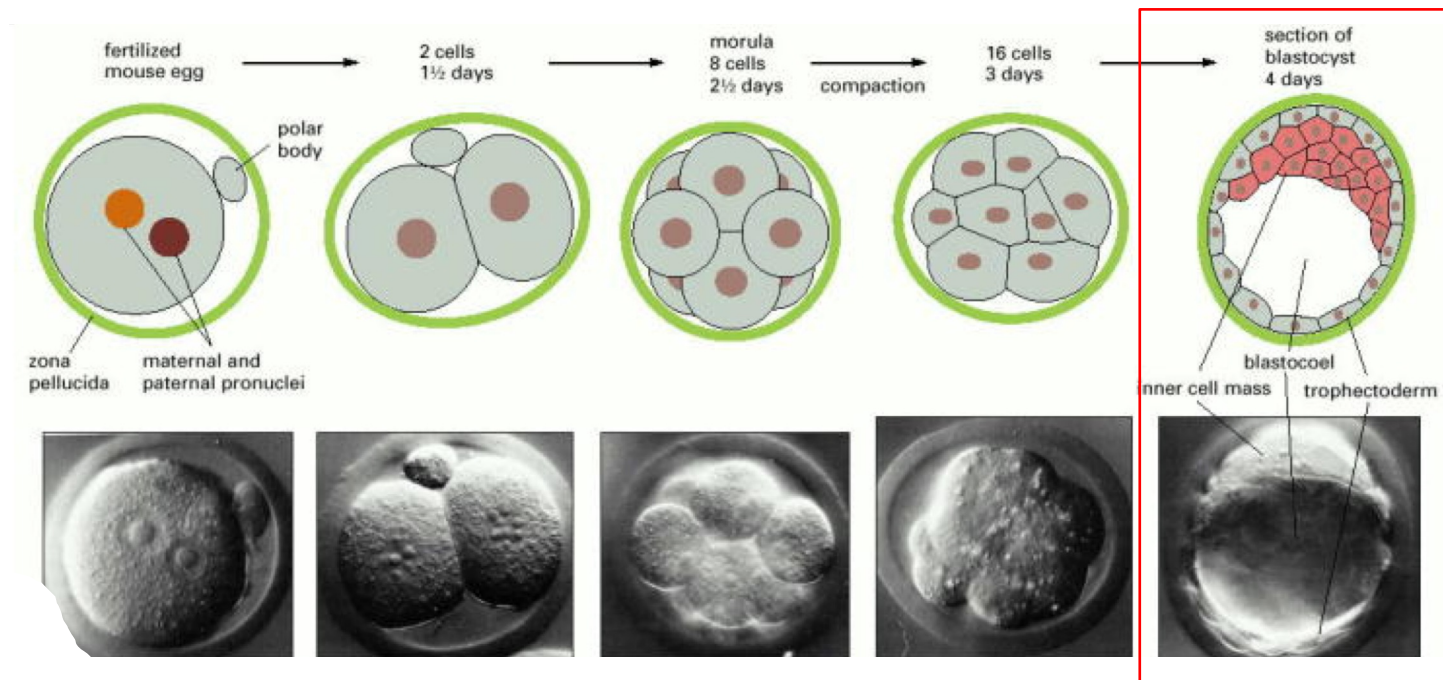
Site-specific Recombination:

A specific sequence is recognized and recombined by an enzyme.

Homologous recombination : targeted integration of the transgene into the host genome

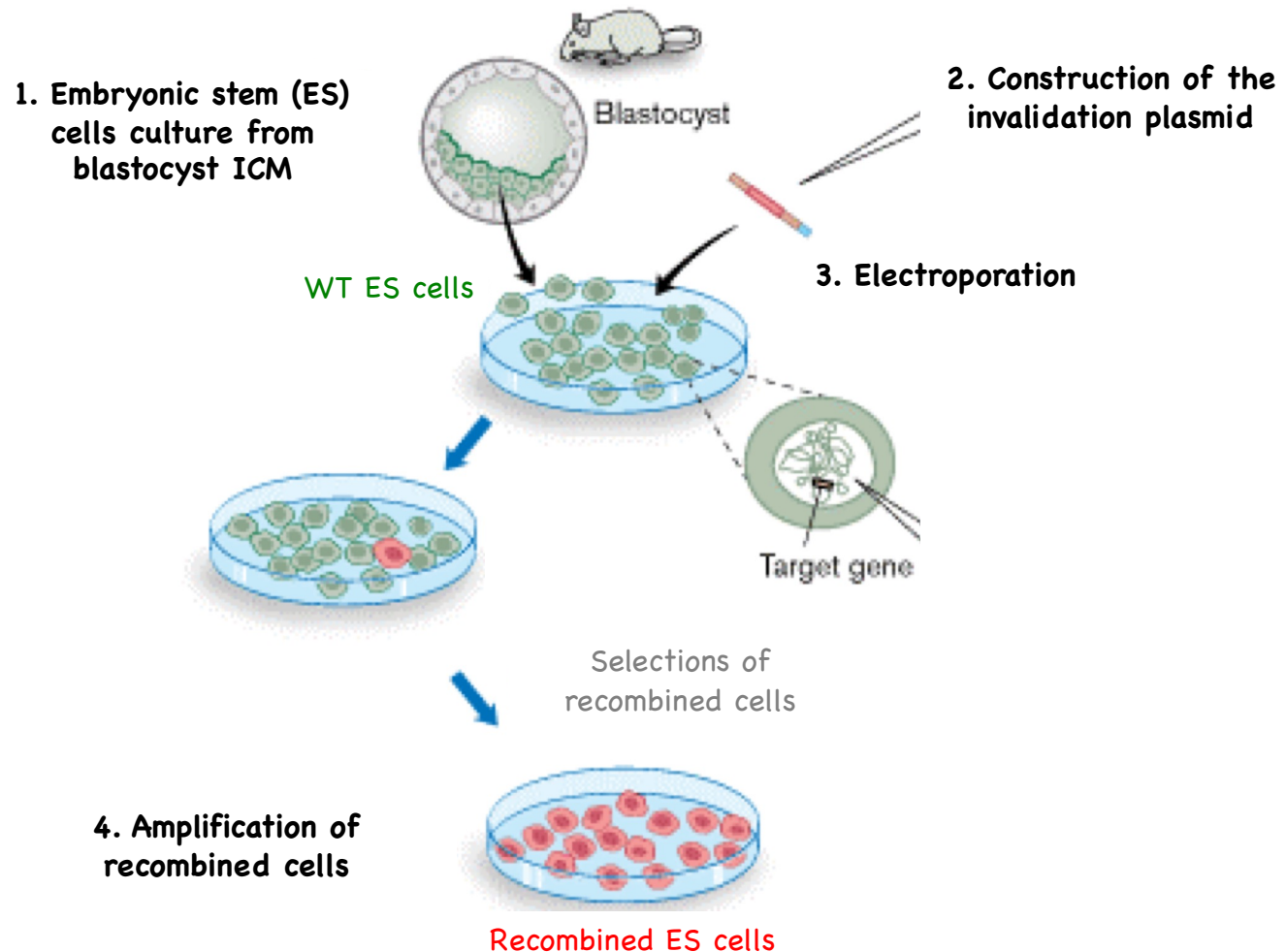
General strategy used for the production of KO mice by homologous recombination:

1. Embryonic stem cells culture

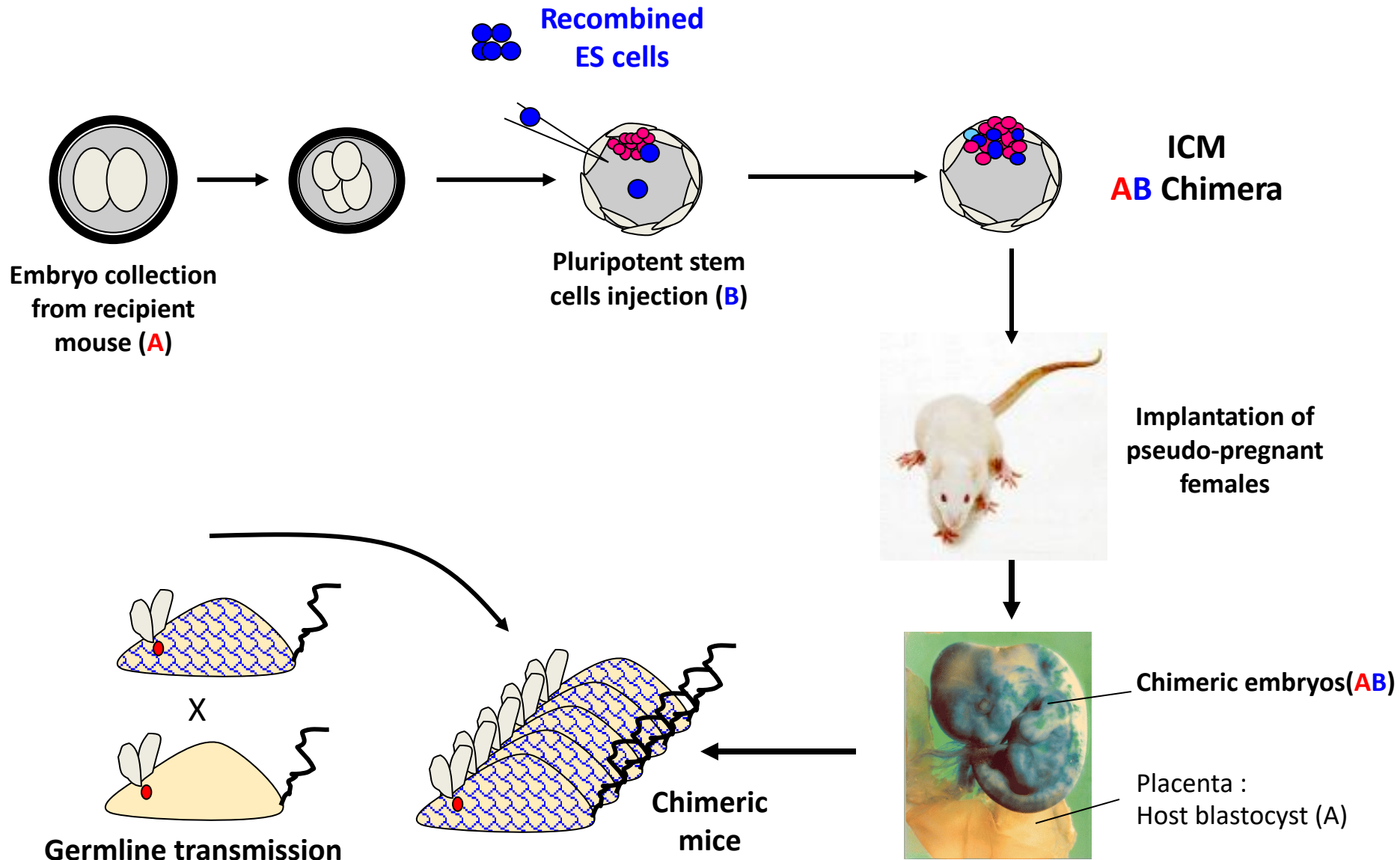


Homologous recombination : targeted integration of the transgene into the host genome

General strategy for the production of KO mice by homologous recombination:



Homologous recombination : targeted integration of the transgene into the host genome

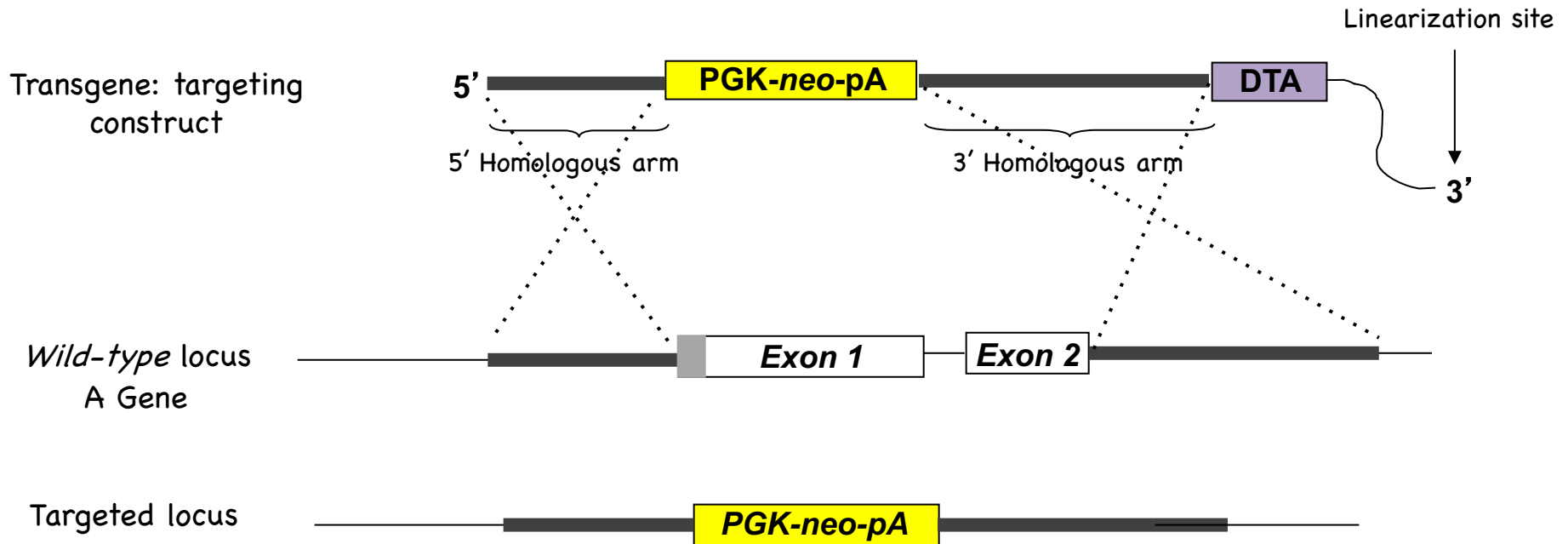


Homologous recombination : targeted integration of the transgene into the host genome

Construction of the invalidation plasmid: different strategies depending on the type of recombination.

Homologous recombination : targeted integration of the transgene into the host genome

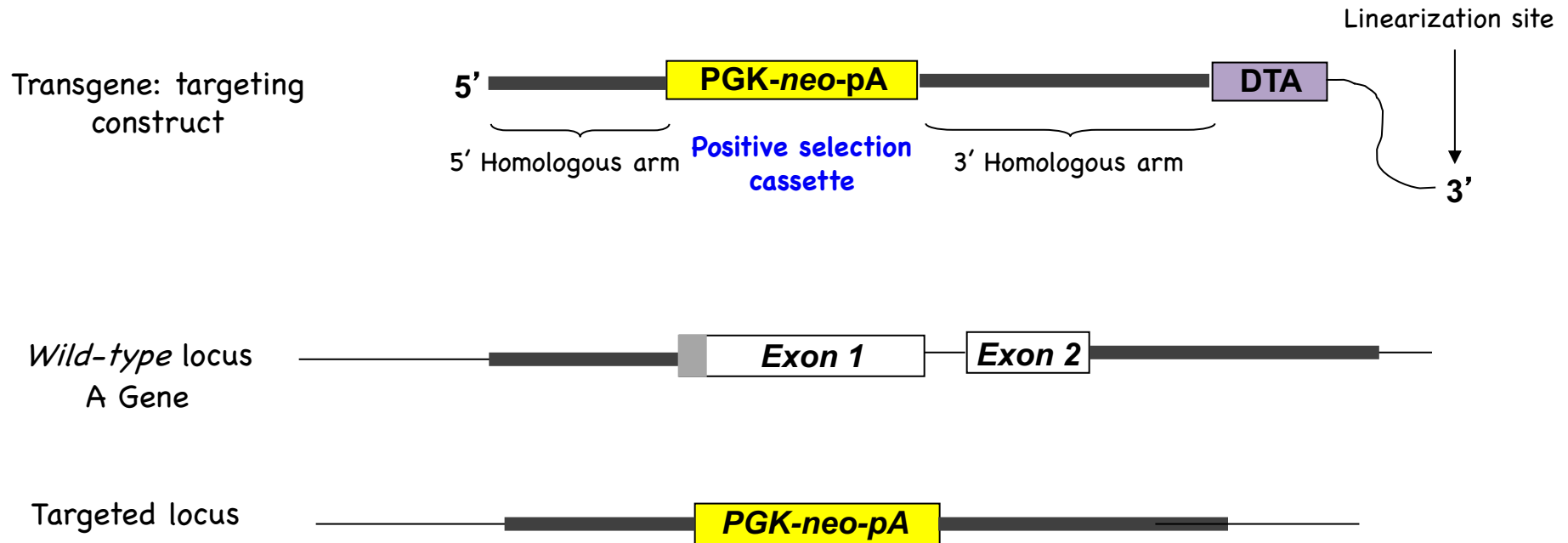
Global recombination : complete knock-out



Homologous arm: homologous sequences to those of the target gene.

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out



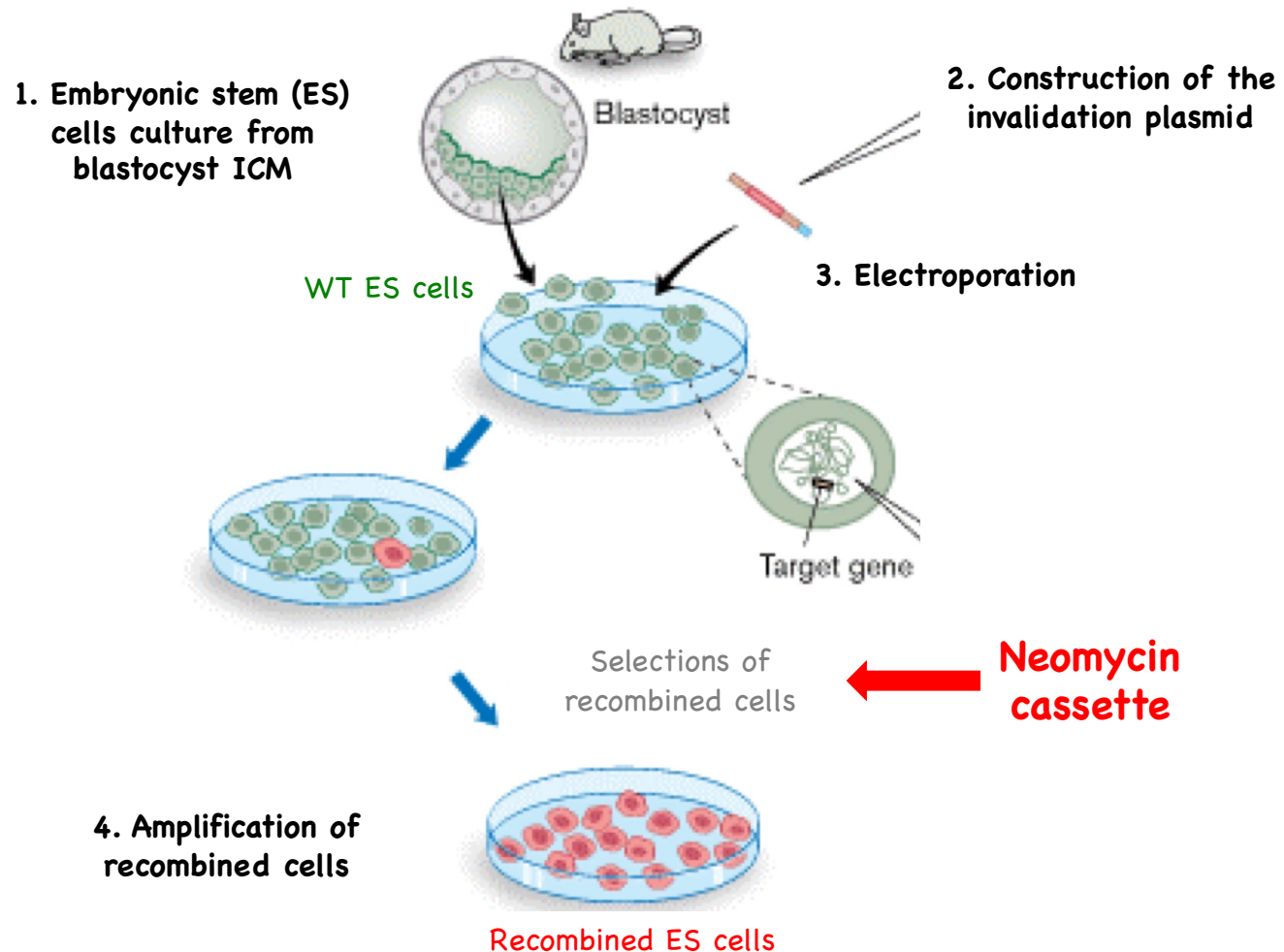
Positive selection Cassette (localized into the homologous region)

PGK: promoter with high activity in targeted cells (ES cells).

Neo: *Neomycin*, resistance gene to G418 antibiotic.

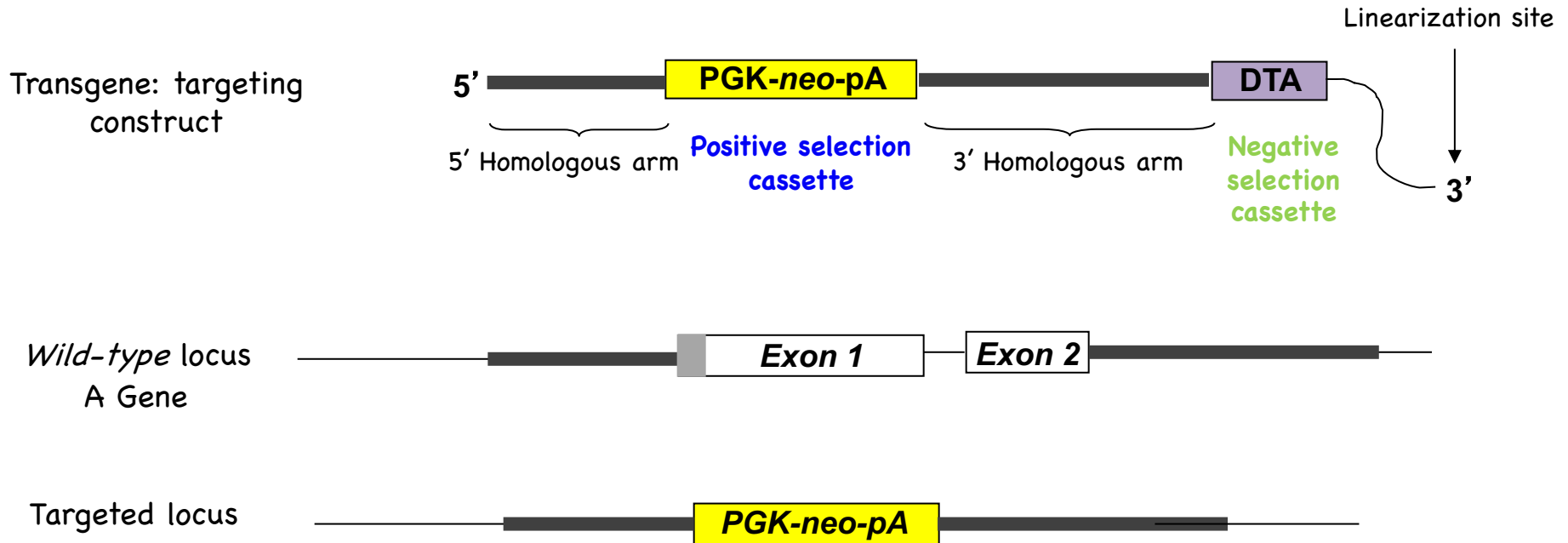
Homologous recombination : targeted integration of the transgene into the host genome

General strategy for the production of KO mice by homologous recombination:



Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out



Positive selection Cassette (localized into the homologous region)

PGK: promoter with high activity in targeted cells (ES cells).

Neo: Neomycine, resistance gene to G418 antibiotic.

Negative selection cassette (localized out of the homologous region)

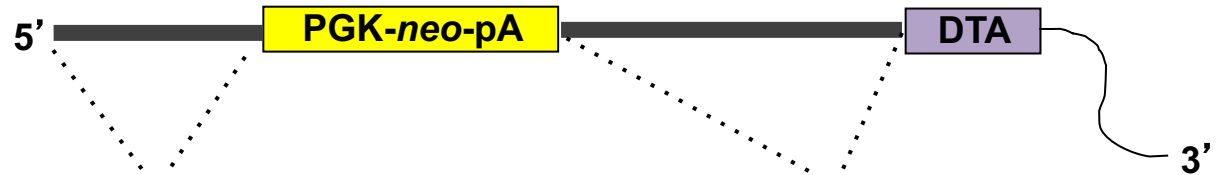
HSV-*tk*: HSV-thymidine kinase (resistance to ganciclovir antibiotic)

DT-A: diphtheria toxin A (when expressed, the toxin leads to cell death)

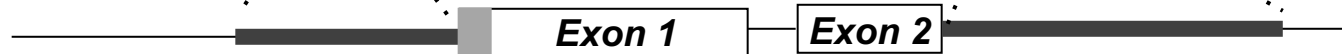
Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

Transgene: targeting construct



Locus wild-type
A Gene



Targeted locus

A^{+/-} ou A^{neo/+}



Positive selection Cassette (localized into the homologous region)

PGK: promoter with high activity in targeted cells (ES cells).

Neo: Neomycine, resistance gene to G418 antibiotic.

Negative selection cassette (localized out of the homologous region)

HSV-*tk*: HSV-thymidine kinase (resistance to ganciclovir antibiotic)

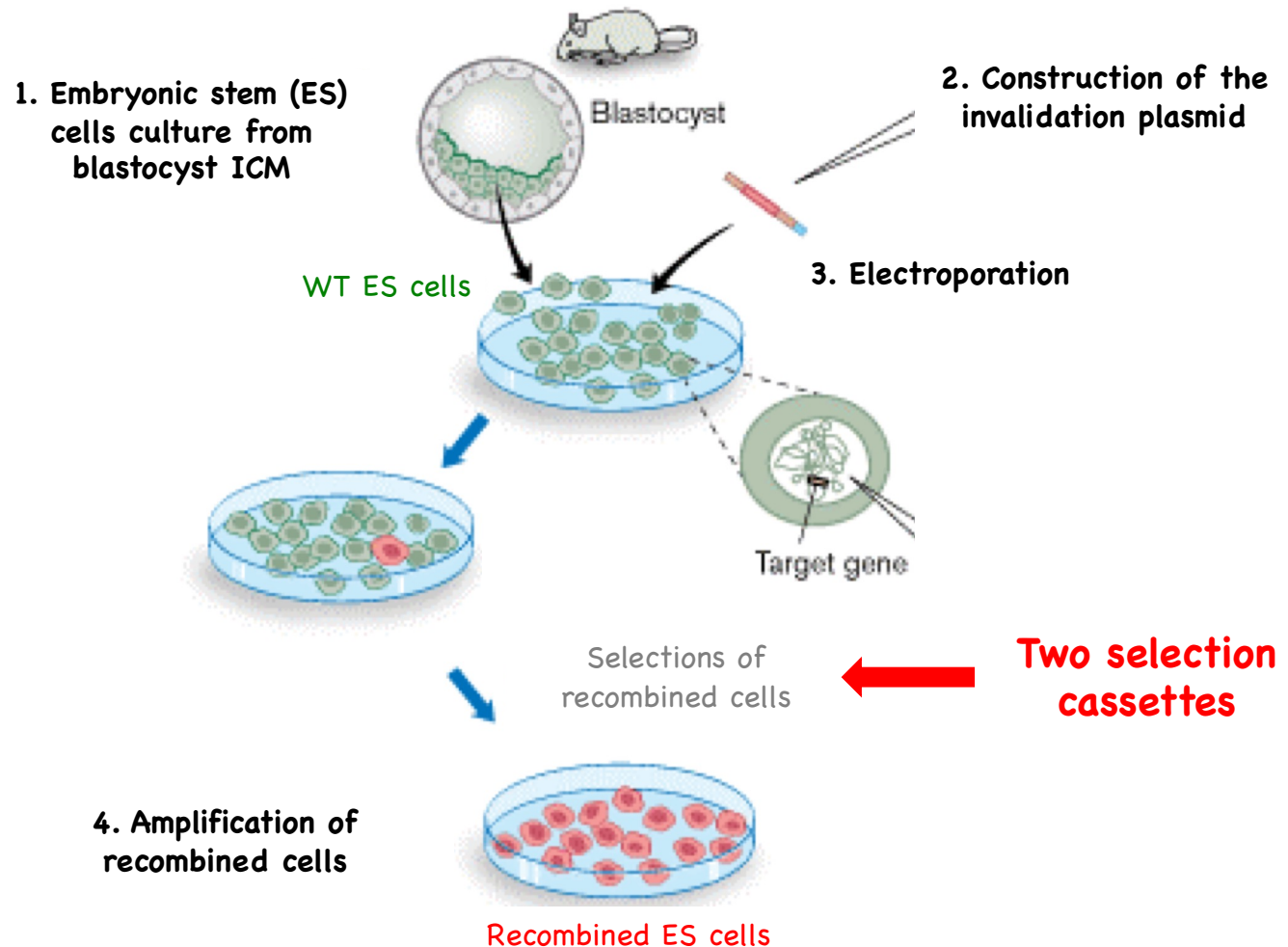
DT-A: diphtheria toxin A (when expressed, the toxin leads to cell death)

Recombined cells

- 1.resistant to G418 (*neomycin*⁺)
- 2.resistant to diphtheria toxin or to ganciclovir

Homologous recombination : targeted integration of the transgene into the host genome

General strategy for the production of KO mice by homologous recombination:



Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

Gene dosage study

For most genes, the loss of one allele does not result in altered expression level thanks to compensation by the second allele.

For some genes, a gene dosage is observed.

Ex: *Pitx3* is expressed in the lens.

- *Pitx3*^{+/+}: 100% of *Pitx3* expression.
- *Pitx3*^{+/-}: Loss of 50% of *Pitx3* expression levels. Normal lens formation, but cataracts appears at 6 months = **Hypomorph**, intermediate phenotype between WT and KO genotypes.
- *Pitx3*^{-/-}: Complete loss of *Pitx3* expression (KO). Defective lens formation.

Homologous recombination : targeted integration of the transgene into the host genome

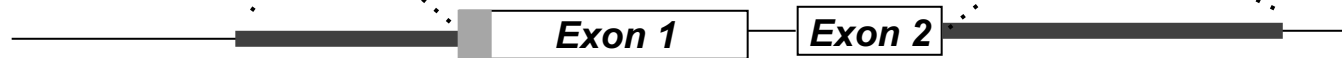
Global recombination : complete knock-out

Rep: reporter gene (ex: LacZ, GFP, RFP)

Transgene: targeting construct



Wild-type locus
A Gene



Targeted locus

A^{Rep/+} ou
A^{Rep-neo/+}



Reporter gene expression permits to follow :

(1) the gene endogenous expression in heterozygous individuals.

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

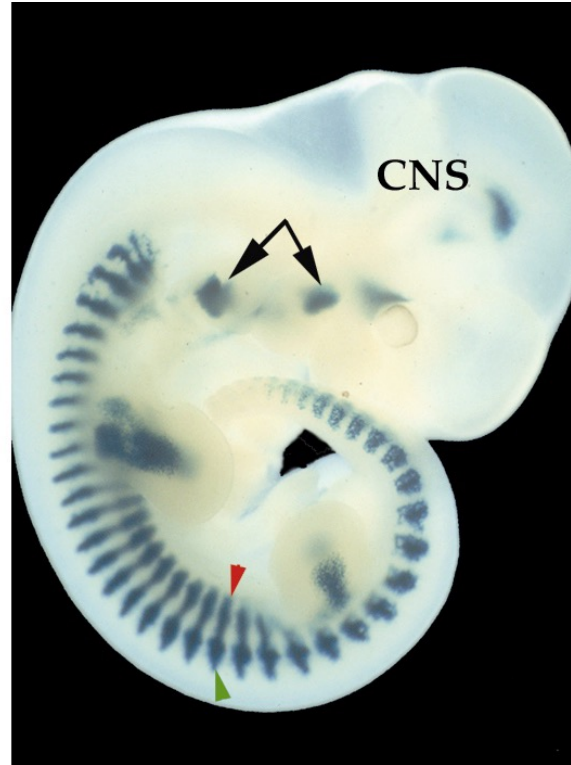
Myf5^{nLacZ/+} embryos

Xgal staining

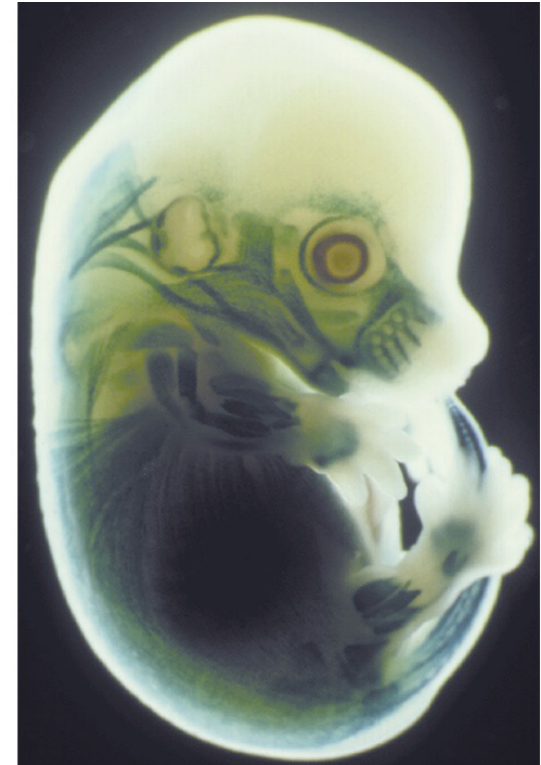
E8.5



E11.5



E14.5



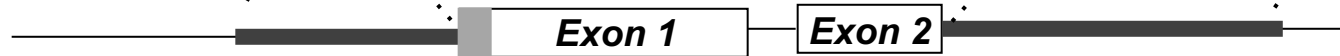
Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

Transgene: targeting construct



Wild-type locus
A Gene



Targeted locus

ARapp/+ ou
ARapp-neo/+

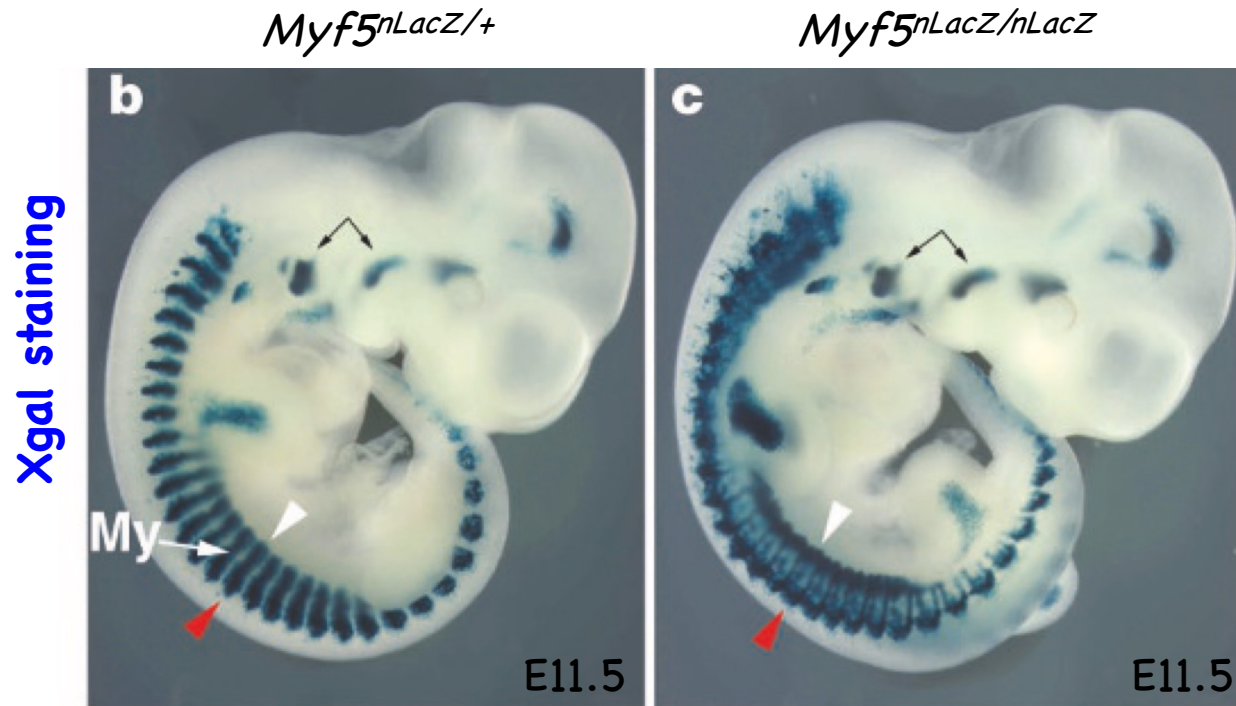


Reporter gene expression permits to follow :

- (1) the gene endogenous expression in heterozygous individuals.
- (2) the fate of mutant cells (KO) in homozygous individuals.

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out



- Analysis and demonstration of *Myf5* role during embryonic myogenesis
- Abnormal localization of *Myf5^{nLacZ}* myogenic precursors

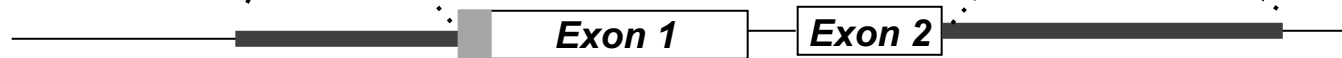
Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

Transgène: targeting construct



Locus wild-type
Gène A



Targeted locus

ARapp/+ ou
ARapp-neo/+



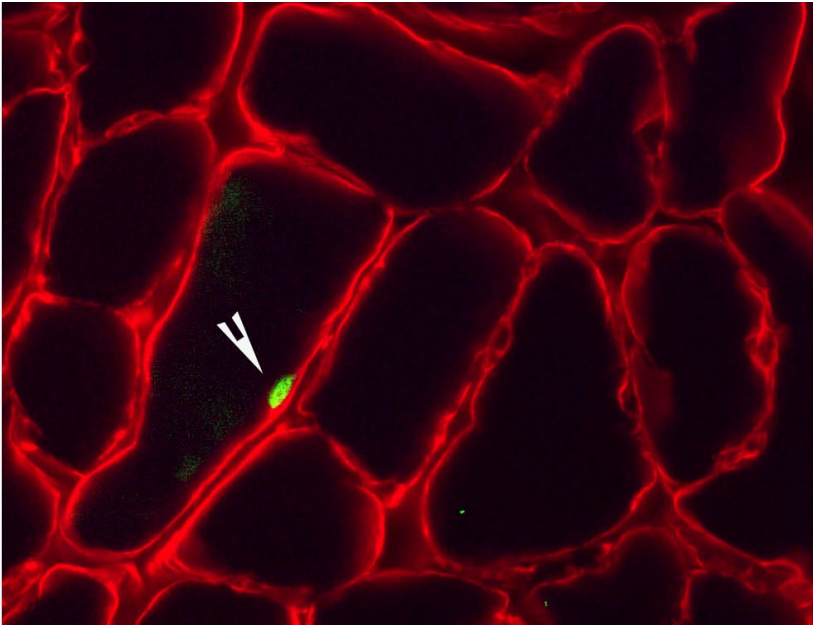
Reporter gene expression permits :

- (1) To determine the gene endogenous expression in heterozygous individuals.
- (2) To follow the fate of mutant cells (KO) in homozygous individuals.
- (3) To purify cells with experimental or therapeutic interest

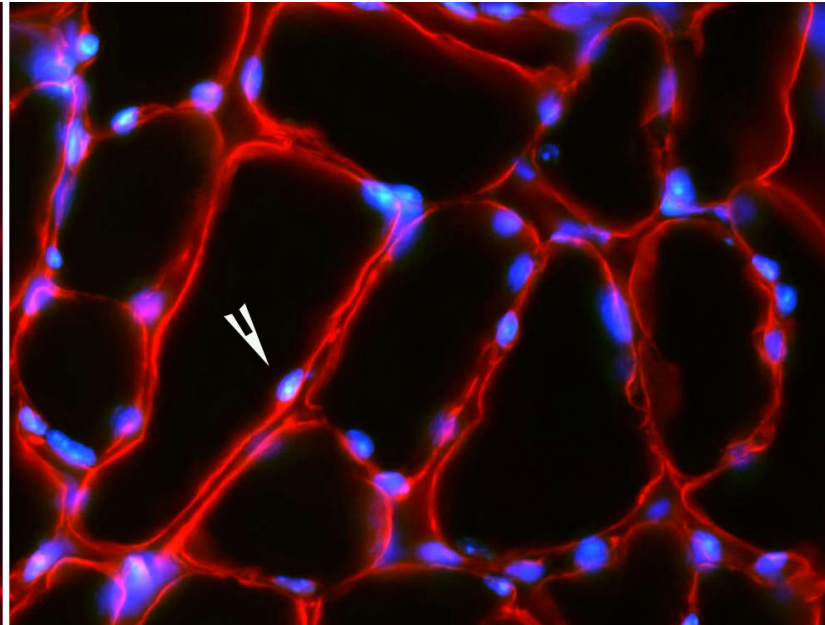
Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

Laminin GFP staining



Laminin DAPI staining



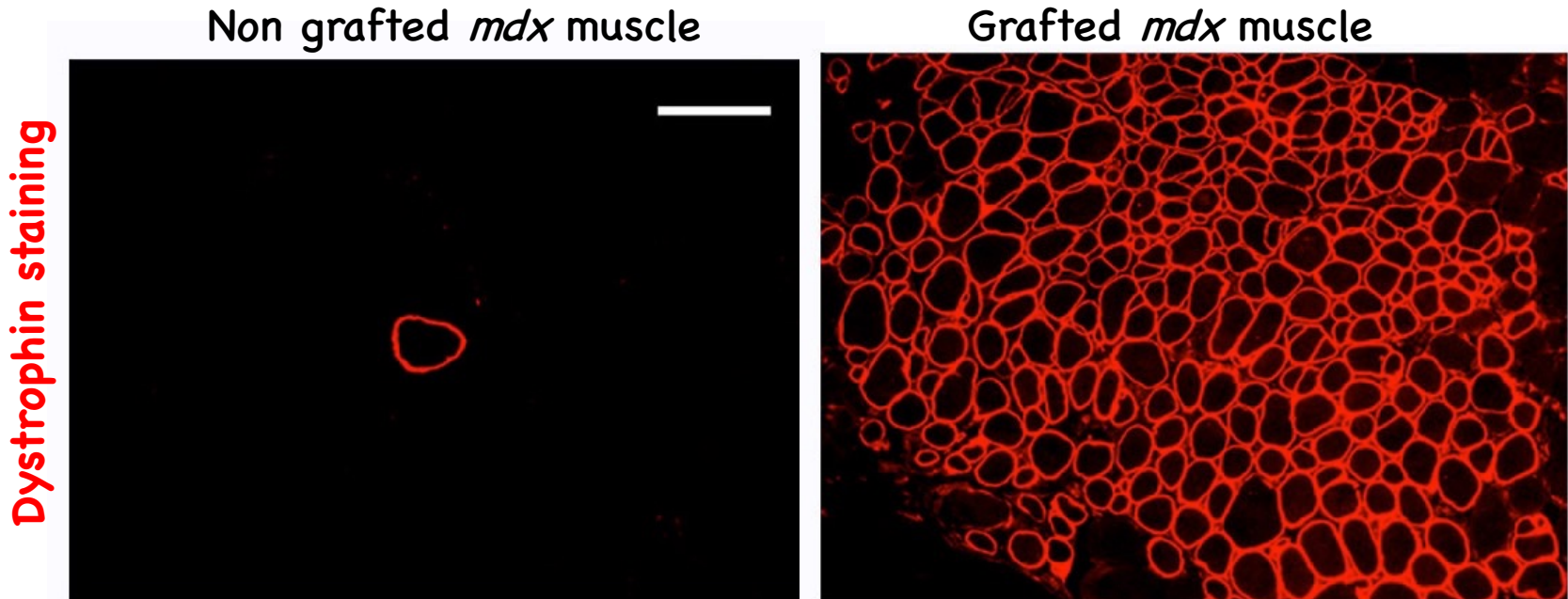
Skeletal muscle section of adult *Pax3^{GFP/+}* mice

Pax3 gene is expressed in skeletal muscle stem cells.

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

After purification by flow cytometry, Pax3(GFP+) muscle stem cells are grafted into *mdx* dystrophic mice skeletal muscle.

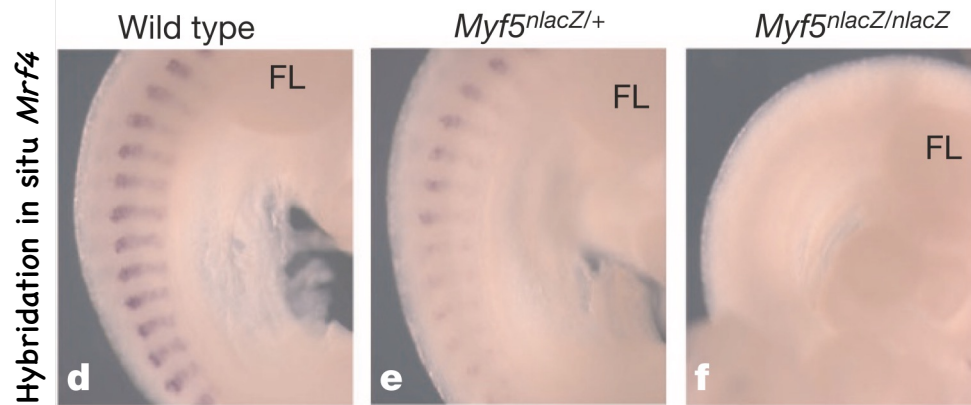
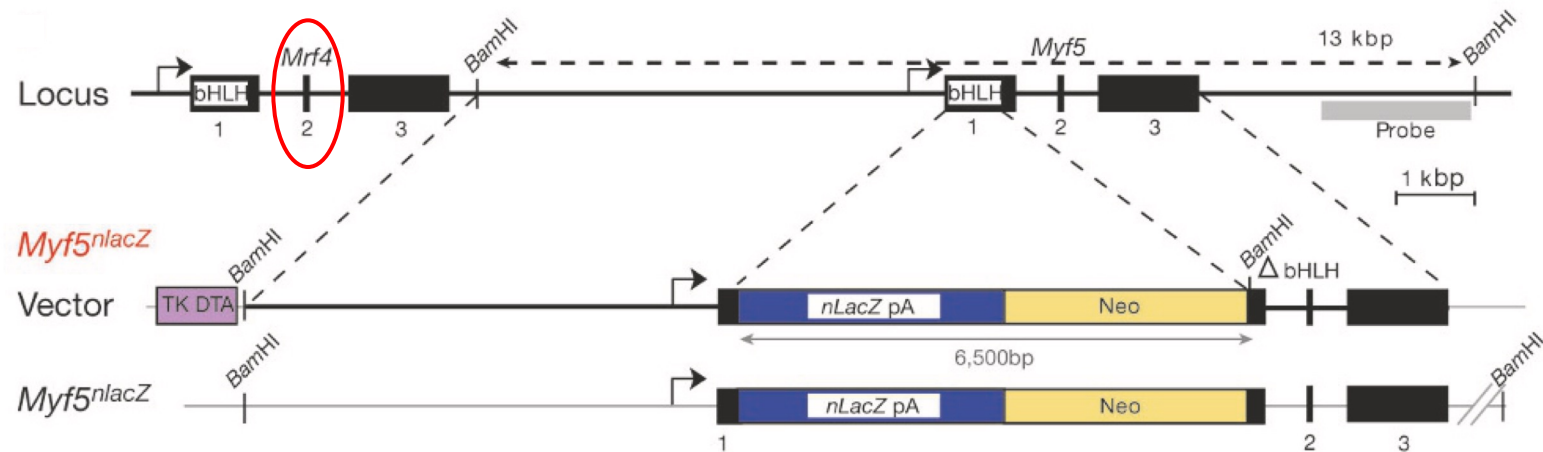


Rescue of dystrophin expression through grafting

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

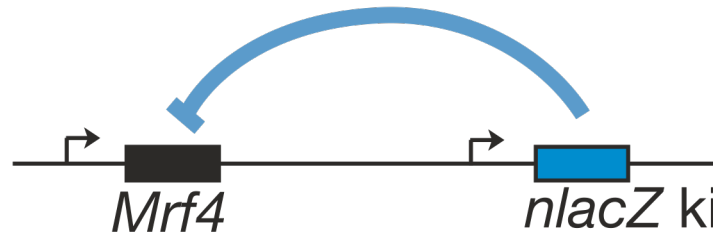
The hazards of loci organization



Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out

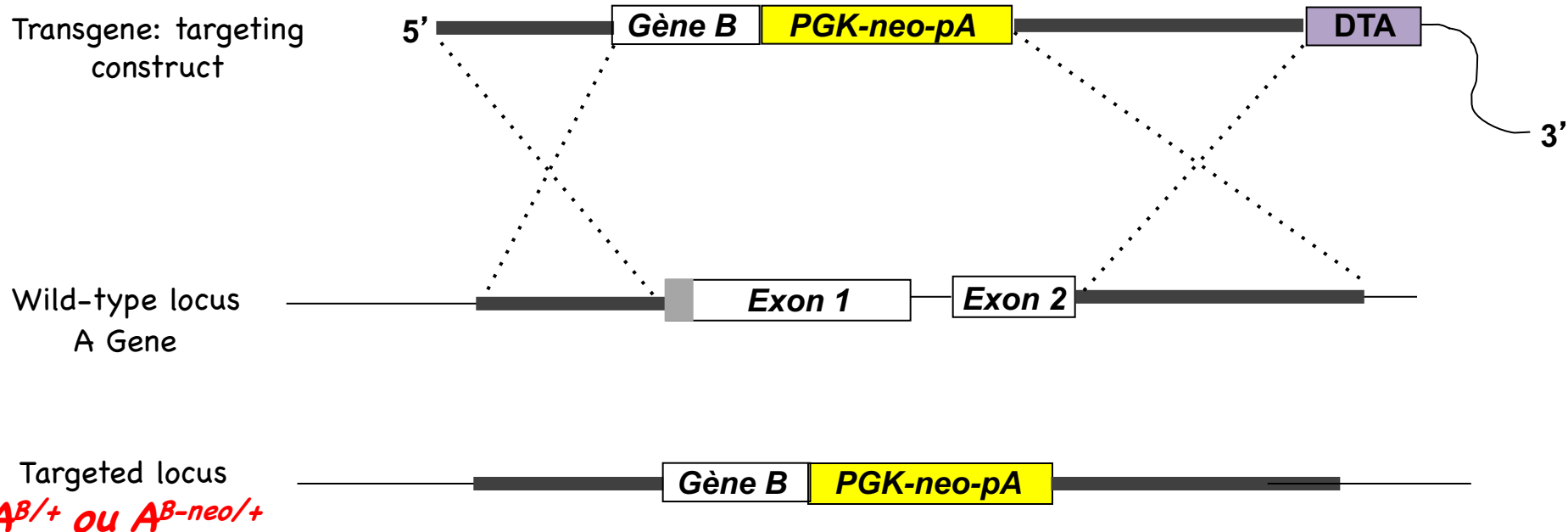
The hazards of loci organization



In fact, *Myf5^{nlacZ/nlacZ}* is a double mutant *Myf5^{-/-}, Mrf4^{-/-}*

Homologous recombination : targeted integration of the transgene into the host genome

Global recombination : complete knock-out



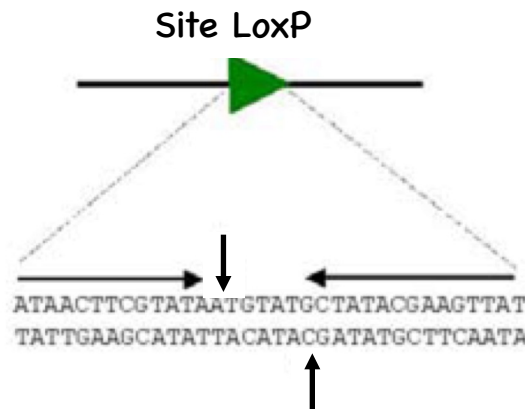
It is possible to replace endogenous gene expression by that of another gene in order to study their functional redundancy.

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out

Use of Cre recombinase (cyclic):

- 38 Kd protein expressed by P1Lox bacteriophage.
- Can catalyse the recombination between two Lox sites.
- Lox sites are 34bp sequences composed by 13bp palindromes at each end, separated by a conserved 8bp sequence.
- Lox sites do not exist into the eukaryotic genome.



Homologous recombination : targeted integration of the transgene into the host genome

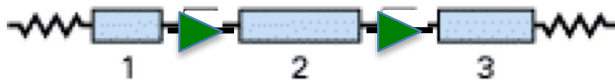
site-specific recombination : conditional knock-out

Use of Cre recombinase (cyclic):

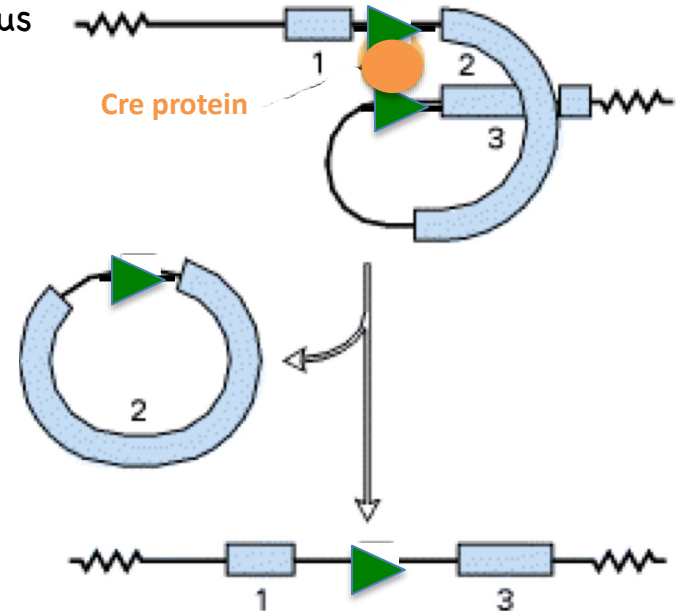
Without Cre

In presence of Cre

Targeted locus
A Gene



Targeted locus
A Gene



Gene invalidation

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out

Use of FLP recombinase :

- Protein expressed by *Saccharomyces cerevisiae*.
- Can catalyse the recombination between two FRT sites.
- Frt sites are 34bp sequences composed by 13bp palindromes at each end, separated by a conserved 8bp sequence.
- FRT sites do not exist into the murine genome.
- FLP recombinase acts similarly to Cre protein.

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out

Transgene: targeting construct



Wild-type locus
A Gene



Targeted locus
A_{flox-neo/+}

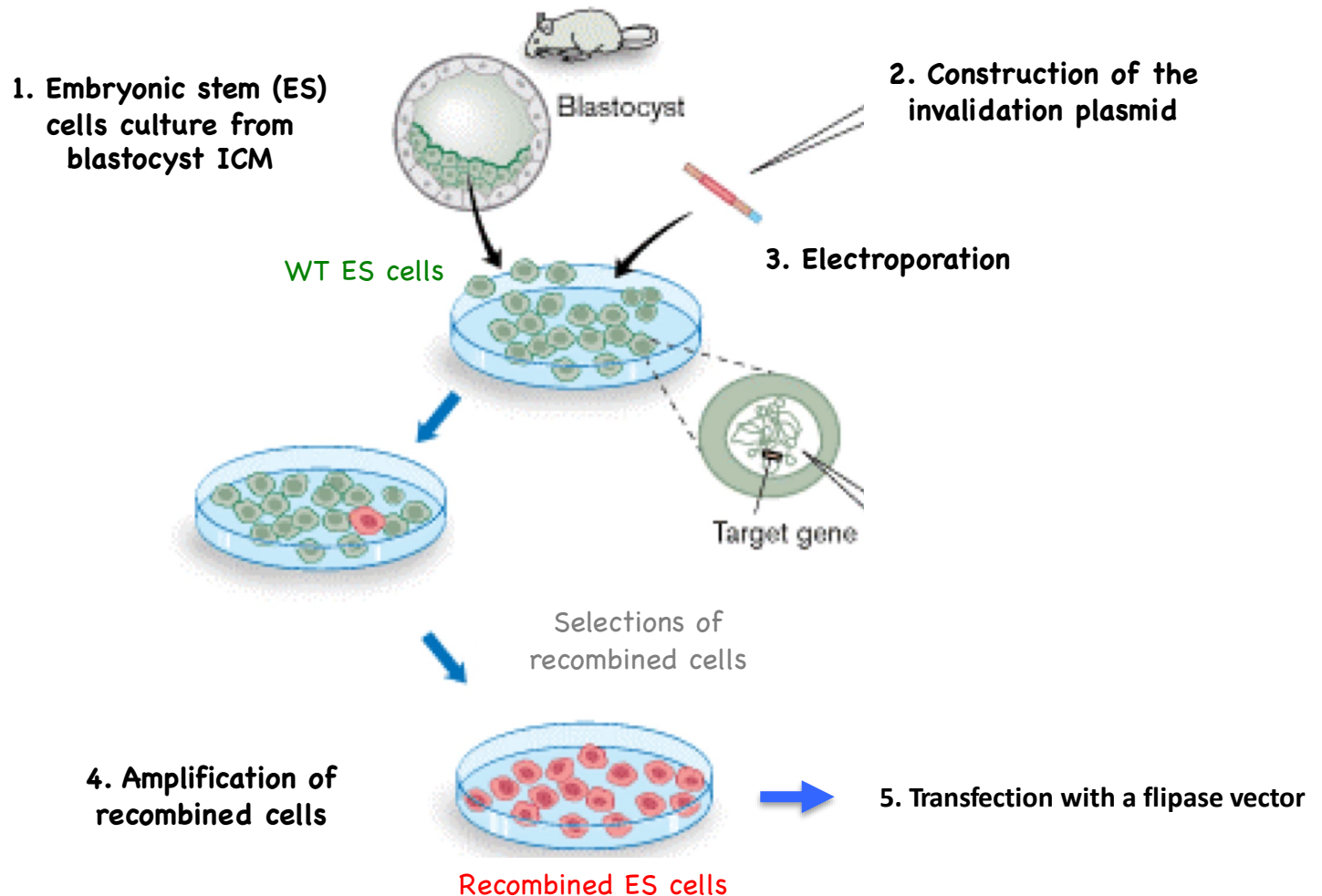


 LoxP sites

 FRT sites

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out



Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out

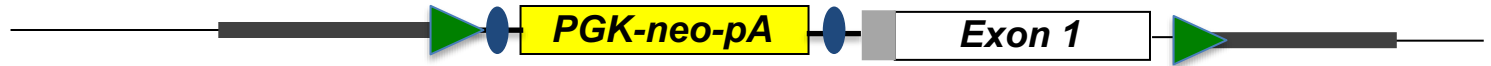
Transgene: targeting construct



Wild-type locus
A Gene



Targeted locus
A Gene



Transfection Flippase

Targeted locus
A^{flox/+}

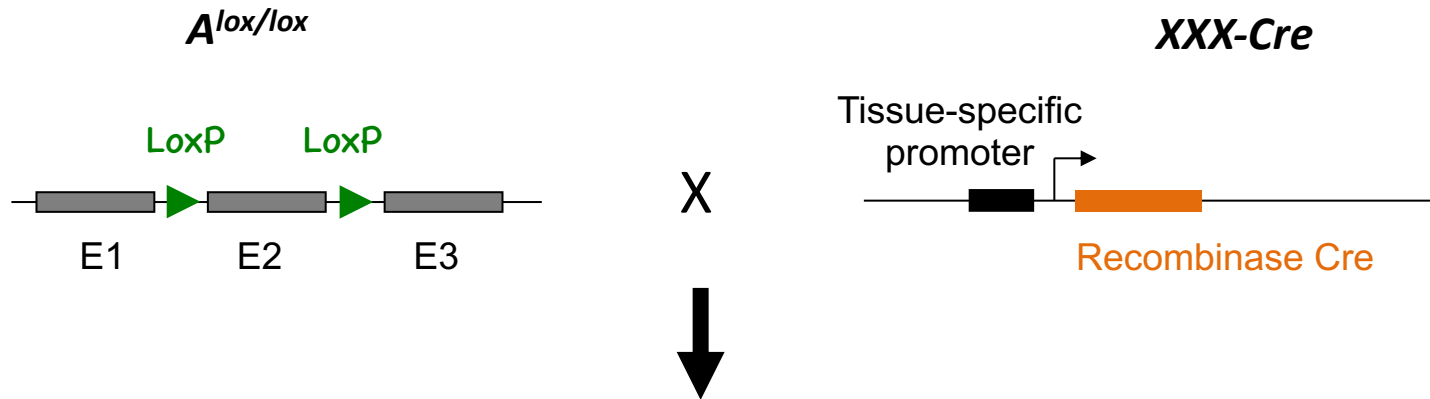


LoxP sites

FRT sites

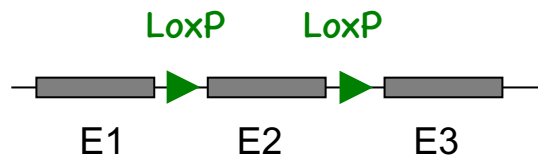
Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : conditional knock-out

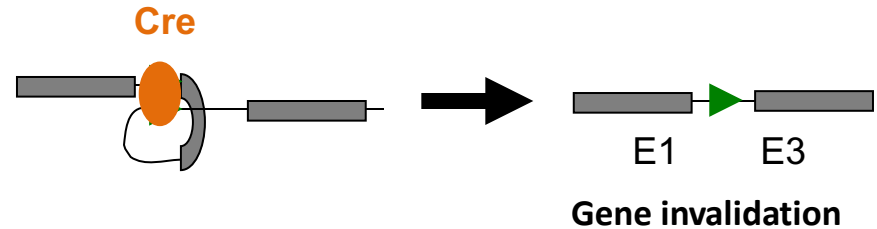


A^{lox/lox}, XXX-Cre

Tissues that do not express Cre recombinase

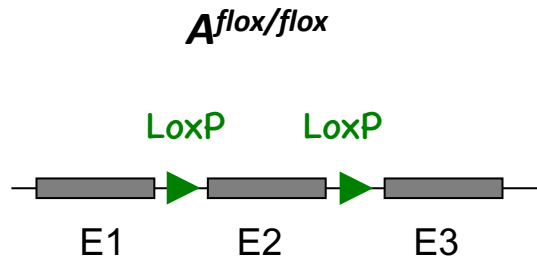


Tissues that express Cre recombinase

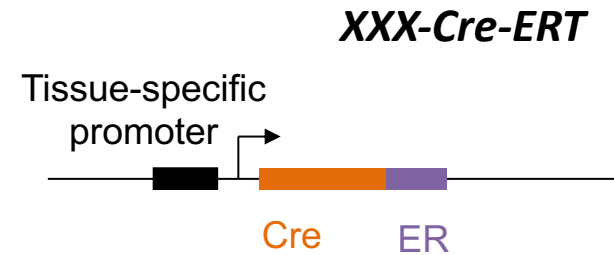


Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : inducible and conditional knock-out



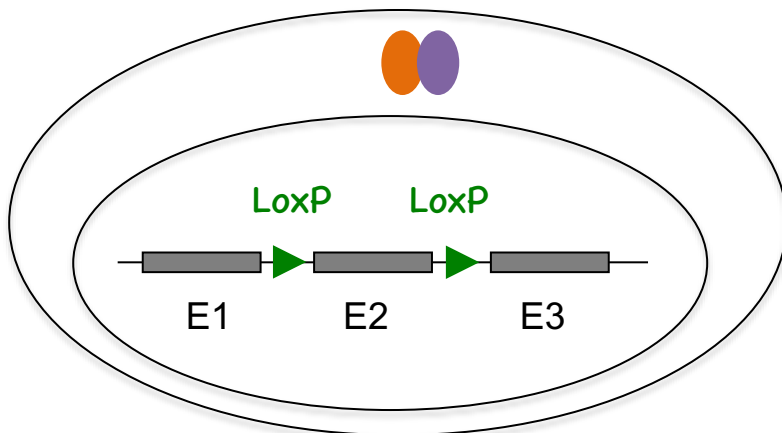
X



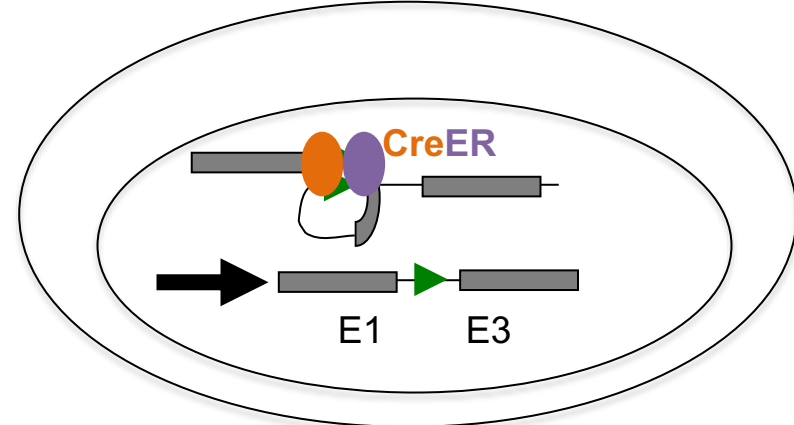
ER: nuclear translocation signal of the Oestrogene Receptor
(Hydroxy-tamoxifen : 4-OHT)

A^{lox/lox},XXX-Cre-ERT

Tissues that express recombinase Cre
in absence of 4-OHT



Tissues that express recombinase Cre
in presence of 4-OHT



CreER nuclear translocation and
gene invalidation

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : inducible and conditional knock-out

Allows gene inactivation in a specific cell type and at a specific time point.

Mice are subjected to intra-peritoneal (IP) injections during 5 consecutive days in order to obtain a 70% to 100% recombination.

Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : inducible and conditional knock-out

Transgene: targeting construct



Pax7 Locus



Targeted *Pax7* locus



Flippase transfection

Targeted locus

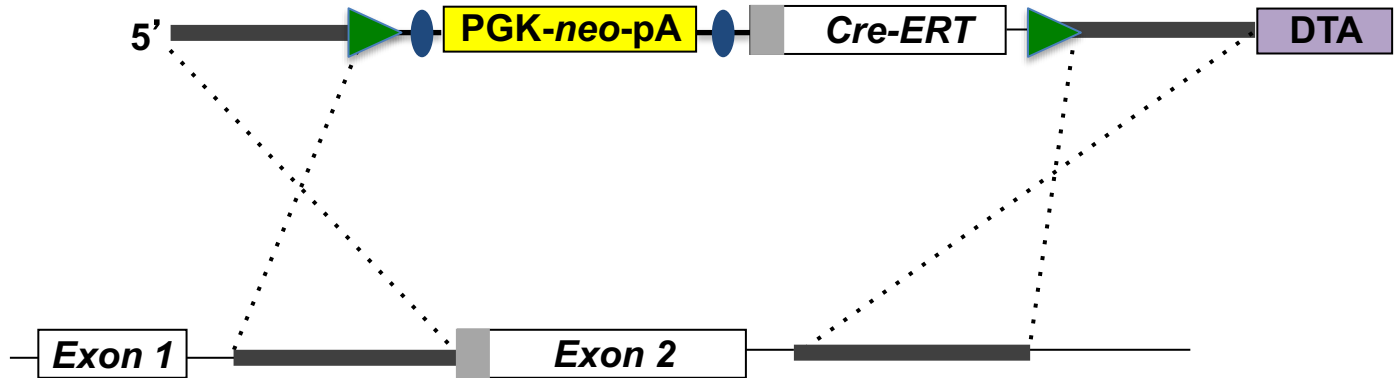
***Pax7*^{Cre-ERT/+}**



Homologous recombination : targeted integration of the transgene into the host genome

site-specific recombination : inducible knock-out

Transgene: targeting construct



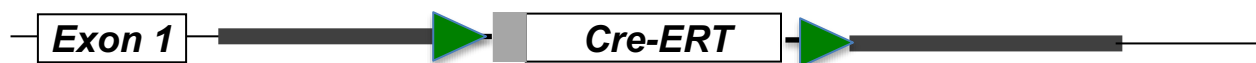
Targeted Rosa locus



Flippase transfection

Targeted locus

R26^{Cre-ERT/+}

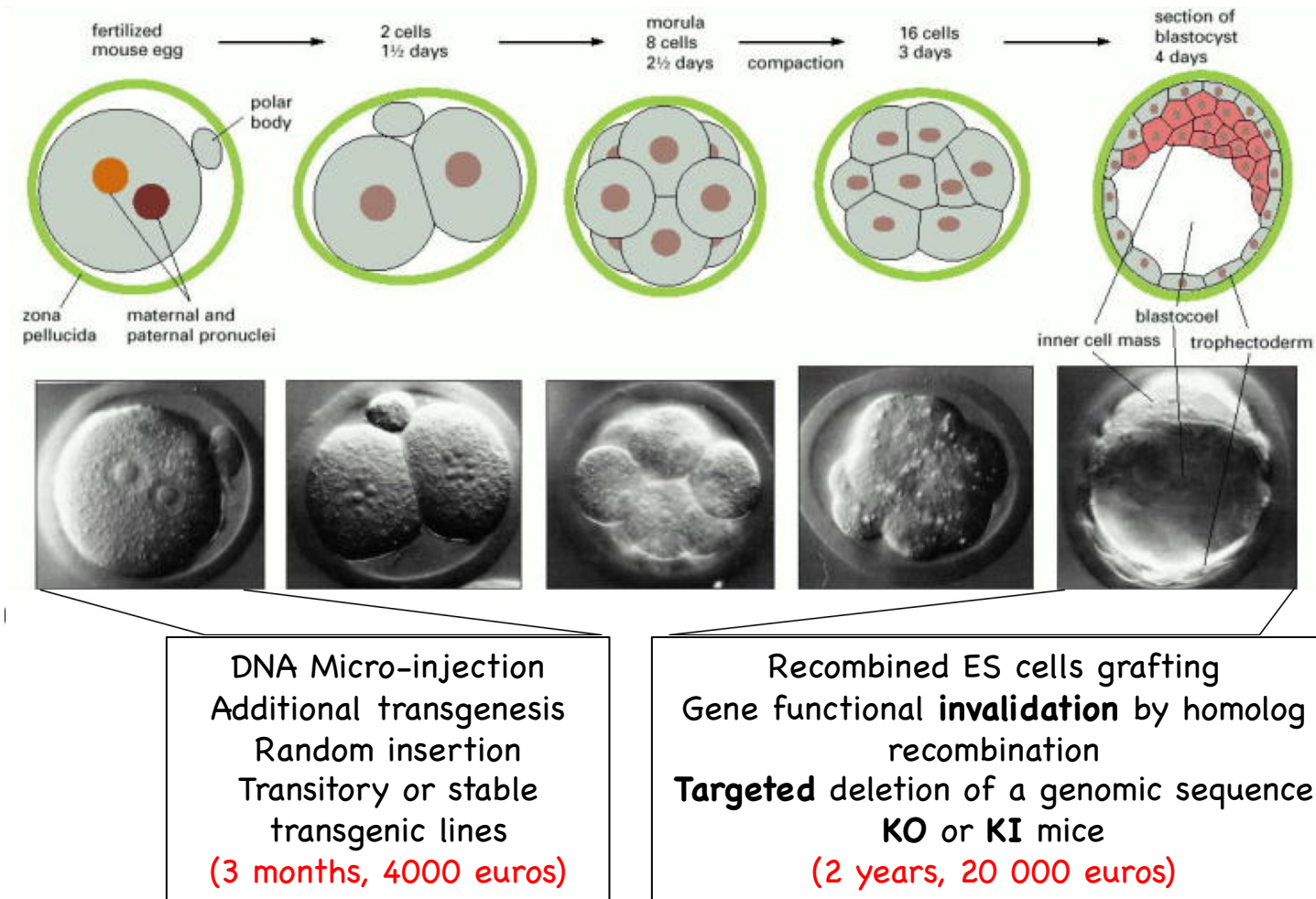


Homologous recombination : targeted integration of the transgene into the host genome

A more complicated technique than classical transgenesis (ES cells culture, cells injection into blastocysts), but much more powerful:

- The expected genetic event is selected in culture: only recombined cells are injected into blastocysts.
- Allows complete gene inactivation or its replacement by another gene (reporter gene, human gene, ...).
- Allows the integration of sequences to control the inactivation of the gene at a given time and/or in a given cell type.

Two different approaches of transgenesis:



A few notions of nomenclature

Hemizygote: only one copy of the indicated gene is present into the genome.

Ex: sexual chromosomes and classical transgenesis by insertion.

Heterozygote: while a mutated copy of the indicated gene is present on the first chromosome, the second one is characterized by a wt copy.

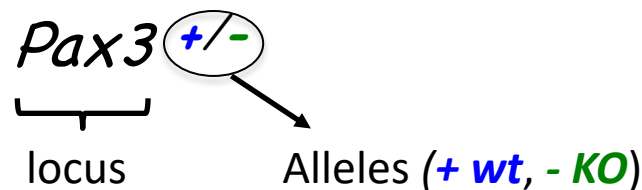
Ex: KO, KI.

Identification of a transgene by insertion: no indicated locus as a consequence of the random insertion. No indication of the transgene number of copies.

Ex: *Pax7-Cre*, *MyoD-Cre-ERT*.

KO or KI identification: targeted insertion into a specific locus. Both alleles states are identified.

Ex: *Pax3*^{+/-}, *Pax7*^{GFP/+}, *Pitx2*^{flox/flox}, *R26R*^{Cre-ERT/+}.



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DNA and **RNA** are indicated in *italic*, in contrast to proteins.

Ex: Pax3 (protein) versus *Pax3* (gene or RNA).

Murine genes and proteins are written in **tiny letters** (first capital letter), while **human** genes and proteins are written **in capital letters**.

Ex: Pitx2 (murine) versus PITX2 (human).

A few notions of nomenclature

Pitx2^{flox/flox}, Pitx3^{flox/flox}, HSA-Cre :

HSA: Human skeletal actin regulatory regions

Pitx2 and *Pitx3* genes conditional inactivation into cells that express Cre recombinase under the control of *Human skeletal actin* gene regulatory regions.

Pitx2^{flox/flox}, Pitx3^{flox/flox}, R26R^{Cre-ERT/+}, Pax3^{GFP/+}

Ubiquitous and 4-OHT inducible *Pitx2* and *Pitx3* inactivation, in a genetic background in which *Pax3*-positive cells express GFP.

Pitx2^{flox/flox}, Pitx3^{-/-}, Pax7^{Cre-ERT/+}

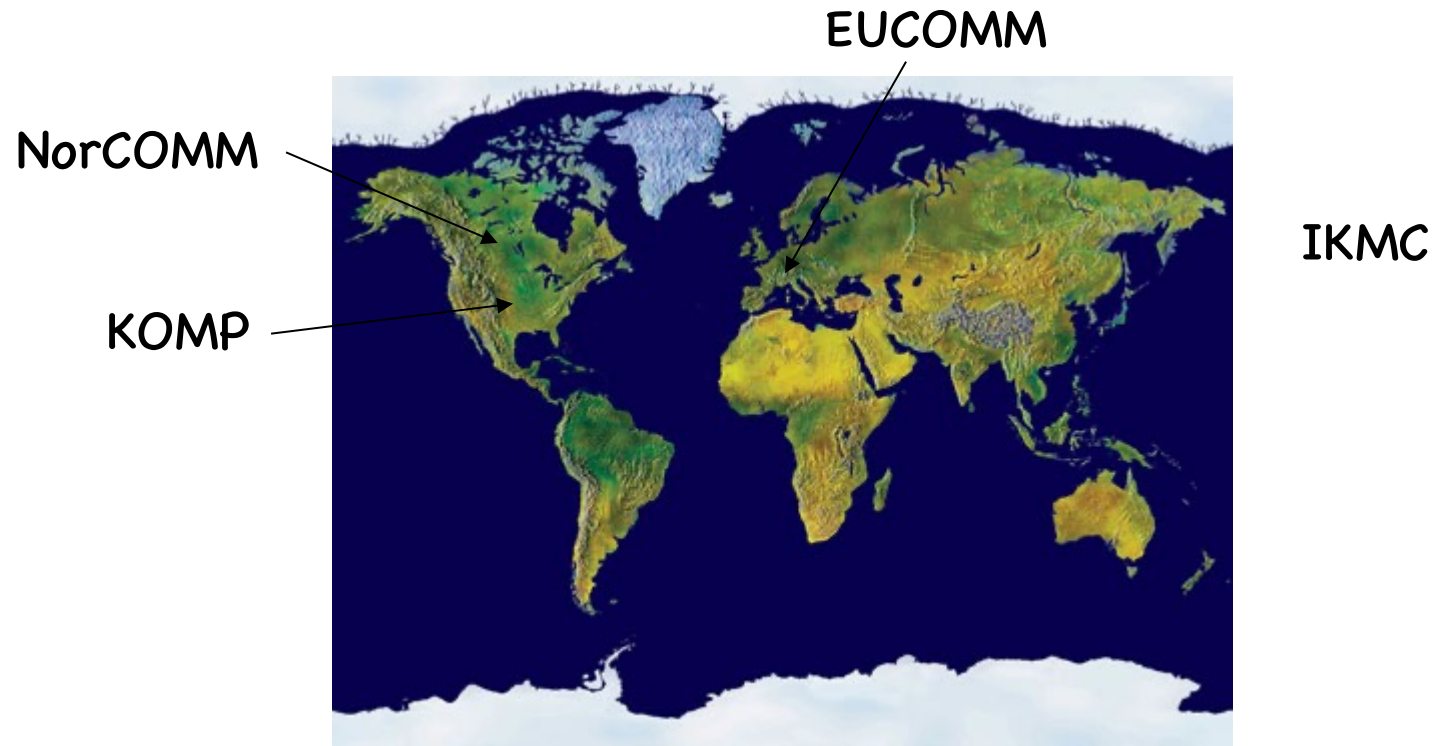
Complete *Pitx3* gene inactivation together with conditional and OHT-inducible *Pitx2* gene inactivation in cells that express the Cre recombinase under the control of *Pax7* gene regulatory regions.

Role of genetic background in the analysis of murine phenotypes

<i>Leptin</i> ^{ob/ob}	C57BL/6 obesity	C57BL/Ks diabetes
<i>APC</i> ^{Min/+}	(B6 x AKR)F1 6 polypes	C57BL/6 30 intestinal tumors
<i>Egf-R</i> ^{-/-}	CD-1 Letal at P21	CF1 Letal post egg- implantation

Production and maintenance of mice transgenic, KO and KI lines

Several research programs have been created in order to target all murine genes, and to conserve murine transgenic lines.



EUCOMM: european Conditional Mouse Mutagenesis Program

KOMP: Knockout Mouse Project (NIH)

NorCOMM (North American Conditional Mouse Mutagenesis project)

IKMC: International Mouse consortium. www.knockoutmouse.org